

Species-Aware Review: Cytochrome *c* Maturation System I in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module:** Bacterial cytochrome *c* maturation (Ccm) System I — covalent heme attachment to exported apo-*c*-cytochromes **Commissioned bucket:** kegg:ppu02010 "ABC transporters" (207 candidate genes) **Iteration:** 3 of 3 (final)

1. Executive summary

P. putida KT2440 encodes a **complete, canonical, single-operon cytochrome *c* maturation System I** (`ccmABCDEFGH` + `ccmI/cycH`) at a contiguous chromosomal locus, **PP_4327 → PP_5748 → PP_4320**. **Every step of the generic module is COVERED** by a dedicated gene with matching domain architecture, membrane topology, and intact catalytic residues.

The most important curation message is a **framing error, not a biology gap**: the commissioned candidate list (KEGG map ppu02010, "ABC transporters") **does not contain the Ccm machinery at all**. The bucket resolved to ABC transporters only because CcmA/CcmB constitute an ABC-family energy module; the 207 listed genes are unrelated importers/exporters (Met, Zn, taurine, dipeptide, LPS, etc.). The module must be **re-anchored to the PP_4320–PP_4327 operon**.

Two annotation problems require action: (i) KEGG maps **both** PP_4320 and PP_5748 to `ccmH` (K02200), but domains show they are functionally distinct — **PP_5748 = redox CcmH**, **PP_4320 = CcmI/CycH chaperone** — so the apparent `ccmI` (K02201) "gap" is a **false negative**; (ii) several ppu display names are imprecise (e.g., PP_4321 CcmG labelled "holocytochrome *c* synthetase subunit"). All evidence for KT2440 is **homology/prediction-tier** (no strain-specific *ccm* mutant published), but functional need is strong: KT2440's respiratory chain uses a *cbb₃*-type cytochrome *c* oxidase and other *c*-type cytochromes.

2. Target-organism pathway definition

Process included: Post-translational, periplasmic **covalent attachment of heme *b* to the CXXCH motif of Sec-exported apo-c-cytochromes**, via System I: (a) ATP-energized heme handling and holo-CcmE formation (CcmABCDE), (b) thiol-reductive preparation of the apocytochrome CXXCH cysteines (CcmG + CcmH, fed by DsbD), and (c) covalent heme ligation by the CcmF/CcmH synthase, assisted by the CcmI/CycH apocytochrome chaperone.

Kept separate (neighboring processes / boundary): - KEGG **ppu02010 "ABC transporters"** and **ppu00860 "Porphyrin/heme biosynthesis"** — external dependencies, not part of the module. - **Sec translocon (SecYEG)** and **DsbD** — upstream boundary dependencies (heme substrate is made by the porphyrin pathway; apoprotein export by Sec; reducing equivalents by DsbD → CcmG). - **Cytochrome *c* oxidase / oxidative-phosphorylation maps (ppu00190)** — these are *clients* of Ccm, not part of it. - **System II (Ccs/Res, ccsBA)** — an *alternative* maturation system; **absent** in KT2440 (see §5), so it must not be conflated.

Alternate names/definitions: "Cytochrome *c* biogenesis pathway I"; genes historically named *helABCDX* (Rhodobacter), *cycHJKL/ccl* (Ccl1=CcmF, Ccl2=CcmH), *dsbE=ccmG*, *cycH=ccmI*. KEGG module M00259; GO:0017004 (cytochrome complex assembly) / GO:0018063 (cytochrome *c*-heme linkage).

3. Expected step model → gene mapping

Module step (generic)	Expected player	KT2440 gene	KO	Call
CcmA ATPase (energy)	CcmA ATPase	PP_4327	K02193	covered
CcmB membrane subunit	CcmB	PP_4326	K02194	covered
CcmC heme-handling	CcmC	PP_4325	K02195	covered
CcmD accessory	CcmD	PP_4324	K02196	covered
CcmE heme chaperone	CcmE	PP_4323	K02197	covered
CcmG thiol-disulfide oxidoreductase	CcmG/ DsbE	PP_4321	K02199	covered
CcmH redox component	CcmH	PP_5748	K02200	covered
CcmF/CcmH holocytochrome-c synthase	CcmF	PP_4322	K02198	covered
CcmI/CycH accessory	CcmI/CycH	PP_4320	(K02201, mis-mapped to K02200)	covered
<i>External:</i> DsbD electron delivery	DsbD	PP_4235 / PP_0561	K04084	candidate_uncertain (paralog ambiguity)
<i>External:</i> Sec export; heme biosynthesis	—	assumed present	—	out of scope

All nine internal module slots are covered. The operon is fully contiguous (complement strand, ~4.91 Mb): PP_4327→PP_4326→PP_4325→PP_4324→PP_4323→PP_4322→PP_4321→**PP_5748**→PP_4320, i.e. **ccmA-B-C-D-E-F-G-H-I**. PP_5748's out-of-sequence tag is a re-annotation artifact (small ORF inserted in its canonical *ccmH* slot between *ccmG* and *ccmI*), **not** genomic separation.

4. Candidate genes and evidence

Evidence tier for all target-organism assignments: **homology/prediction** (UniProt "Inferred from homology" or "Predicted"; no published KT2440 *ccm* knockout). Domain and motif calls are from InterPro/Pfam + sequence inspection; mechanistic roles transferred from *E. coli* and *Rhodobacter capsulatus* (transfer **strong** given operon synteny and conserved catalytic residues).

- **PP_4327 CcmA** (Q88EX5, 210 aa) — Cytochrome *c* biogenesis ATP-binding export protein, **EC 7.6.2.5**, inner membrane. ABC ATPase energizing heme handling. *Caveat*: EC/ABC annotation is why the gene family enters the ppu02010 bucket — do not treat as a solute transporter.
- **PP_4326 CcmB** (Q88EX6, 222 aa, 6 TM) — Heme exporter B membrane subunit. Covered.
- **PP_4325 CcmC** (Q88EX7, 255 aa, 6 TM) — Heme-handling subunit (WWD/heme-binding, Cyt_c biogenesis family). Covered.
- **PP_4324 CcmD** (Q88EX8, 58 aa, 1 TM) — Small accessory subunit; short length is normal for CcmD (not a fragment). Covered.
- **PP_4323 CcmE** (Q88EX9, 151 aa) — OB-fold heme chaperone (Pfam PF03100); **conserved heme-binding His in HxxxY (HDEKY@124)**, equivalent to *E. coli* His130. Covered; functional.
- **PP_4322 CcmF** (Q88EY0, 662 aa, 15 TM) — Holocytochrome-*c* synthetase (Pfam PF01578 + PF16327); periplasmic **WWD-region tryptophans (WWA@233, WWF@244)** and multiple heme-coordinating His. Covered; functional.
- **PP_4321 CcmG/DsbE** (Q88EY1, 178 aa) — Periplasmic thioredoxin-fold disulfide oxidoreductase (Pfam PF08534 Redoxin, DsbE family IPR004799); **redox CXXC = CPSC@73**. Covered. *Caveat*: ppu/UniProt display name "holocytochrome c synthetase subunit" is imprecise — it is CcmG/DsbE.
- **PP_5748 CcmH (redox)** (A0A140FWM4, 158 aa, 1 TM) — CcmH/CycL/Ccl2/NrfF N-terminal redox domain (Pfam PF03918, IPR005616); **redox CXXC = CPKC@45**. Covered. *Caveat*: co-mapped with PP_4320 to K02200; this is the genuine redox CcmH.
- **PP_4320 CcmI/CycH** (A0A140FWM3, 398 aa, 2 TM) — CcmI (IPR017560) + Ig_CycH (PF23892) + TPR_CcmH_CycH (PF23914) + TPR repeats; **no CXXC** → apocytochrome-binding chaperone, **not** a redox subunit. Covered as the CcmI/CycH accessory. *Caveat*: mis-mapped to K02200; should be K02201.

- **PP_4235 / PP_0561 DsbD** (Q88F64 / Q88QD3, 571/590 aa, 8 TM, EC 1.8.1.8) — Two paralogous thiol:disulfide interchange proteins delivering electrons to CcmG. **External** to the module.
Caveat: which paralog services Ccm is unresolved by homology.

5. Gaps, ambiguities, and likely over-annotations

1. **Bucket/candidate-list mismatch (module_needs_revision at the metadata level).** None of the 207 ppu02010 candidates are Ccm genes; the *ccm* operon is absent from the supplied list. The module was mis-seeded from an ABC-transporter overview map. **Re-anchor to PP_4320–PP_4327.**
2. **K02200 double-mapping (annotation over-propagation).** KEGG assigns *ccmH* to both PP_4320 and PP_5748. Domain evidence splits them cleanly: PP_5748 = redox CcmH (CXXC), PP_4320 = CcmI/CycH (TPR, no CXXC). **Reassign PP_4320 → ccmI/CycH (K02201).**
3. **False `ccmI` (K02201) gap.** `link/ppu K02201` returns nothing, implying a gap; in reality CcmI/CycH is present as PP_4320. **Mark covered, not gap.**
4. **System II is genuinely absent (clean boundary).** No `ccsA/resC` (K07399) or fused `ccsBA` (K12265) in ppu. The single K07400 ("ccsB/resB") hit, **PP_2378, is NfuA** (Fe/S biogenesis; Pfam PF01521/PF01106) — a spurious cross-map. So the CcmF/CcmH ligation step is **not** replaced by a lineage alternative → covered, not candidate_uncertain.
5. **Imprecise ppu/UniProt display names** (PP_4321 "holocytochrome c synthetase subunit"; PP_4325 "protoheme IX reservoir complex subunit"). Cosmetic; the KO/domain assignments are correct.
6. **DsbD paralog ambiguity** (external): PP_4235 vs PP_0561 — flag candidate_uncertain but out-of-module.

6. Module and GO-curation recommendations

Step status calls: - **covered:** CcmA (PP_4327), CcmB (PP_4326), CcmC (PP_4325), CcmD (PP_4324), CcmE (PP_4323), CcmF (PP_4322), CcmG (PP_4321), CcmH-redox (PP_5748), CcmI/CycH (PP_4320) — the entire internal module. - **candidate_uncertain:** DsbD electron delivery (PP_4235 / PP_0561) — external boundary, paralog choice unresolved. -

not_expected_in_target_taxon: System-II (Ccs) alternative — correctly absent; do not open a Ccs module for KT2440. - **module_needs_revision (metadata seeding):** replace the ppu02010 ABC-transporter candidate set with the PP_4320–PP_4327 operon.

GO/annotation actions: - Assign **GO:0017004** (cytochrome complex assembly) and **GO:0018063** (cytochrome-c-heme linkage) to the operon members; **GO:0020037** heme binding to CcmC/CcmE/CcmF; **GO:0015232** heme transmembrane transport to CcmABC. - Correct KO mapping: PP_4320 → K02201 (ccmI/cycH); retain PP_5748 → K02200 (ccmH). - No new GO term requests appear necessary; existing terms cover all steps. The **generic module boundary is correct** for this organism (single-operon System I); only the *candidate-gene seeding* was wrong.

7. Genes to promote to full fetch-gene review

1. **PP_4320** — resolve KO/name conflict (CcmI/CycH vs CcmH); confirm reassignment to K02201.
2. **PP_5748** — confirm redox CcmH identity and in-operon position; reconcile the non-sequential locus tag.
3. **PP_4235 and PP_0561** — determine which DsbD paralog feeds the Ccm/CcmG branch vs general periplasmic disulfide/DsbC recycling.
4. (Optional) **PP_4321** — update display name from "holocytochrome c synthetase subunit" to CcmG/DsbE.

8. Key references & evidence provenance

- **Databases (direct for KT2440):** KEGG GENES/REST (link/ppu, gene coordinates) — KO→locus mapping and operon synteny; UniProt UP000000556 (Q88EX5–Q88EY1, A0A140FWM3, A0A140FWM4, Q88F64, Q88QD3) — protein names, TM topology, EC; InterPro/Pfam domains and sequence motifs (CXXC, HxxxY, WWD).
- **Sanders C, Deshmukh M, Astor H, Kranz RG, Daldal F (2005) *J Bacteriol* 187:4245–56.**

P 15937187

CcmI/CycH is an apo-Cyt c chaperone distinct from the CcmF–CcmH(Ccl2) redox/ligation

components — supports the PP_4320 (CcmI) vs PP_5748 (CcmH) split. *Transfer from R. capsulatus: strong (domain-conserved).*

- **Fernández-Piñar R, et al. (2008)** *J Bacteriol* 190:7666–74.

P 18820016

KT2440 respiratory chain includes a *cbb₃* cytochrome *c* oxidase and cytochrome-*c*-related proteins under RoxS/RoxR — establishes functional need for *c*-type cytochrome maturation. *Direct for KT2440.*

- **Grether J, et al. (2025)** *Microb Cell Fact.*

P 39827133

Oxygen-sensitive *cbb₃*-type cytochrome *c* oxidase promoter in KT2440 — corroborates *c*-type oxidase expression. *Direct for KT2440.*

Uncertainty statement: No published *P. putida* KT2440 *ccm* deletion/biochemical study was found; all gene-level assignments are homology/prediction-tier. Confidence is nonetheless high because (i) the operon is syntenic and complete, (ii) catalytic residues (CcmE HxxxY, CcmF WWD/His, CcmG/CcmH CXXC) are intact, and (iii) the strain demonstrably makes *c*-type cytochromes. The one true biological ambiguity is external (which DsbD paralog services Ccm), resolvable by a targeted PP_4235/PP_0561 knockout + cytochrome *c* heme-staining assay.